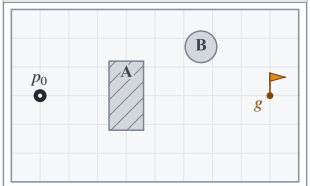
(a) Scene state and task

planner reads exact state s_0

Observation $o_0 = R(s_0)$



$p_0 = (1, 3)$ $g = (9, 3)$

A: $[3.4, 4.6] \times [1.8, 4.2]$

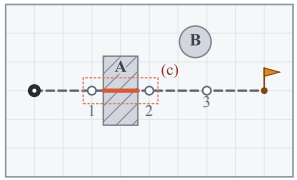
B: $c = (6.6, 4.7)$, $r = 0.55$

Reach g; avoid A and B

(b) Candidate rollouts in one scene

$p_{k+1} = p_k + a_k \Delta t$, $\Delta t = 1$, $\|a_k\|_2 \le 3.5$, $H = 4$; *dashed = simulated*

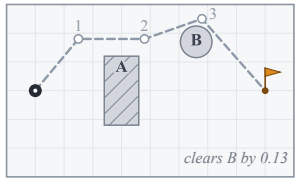
D Direct — collision



k	0	1	2	3
a_x	2	2	2	2
a_y	0	0	0	0

$E = 16.00$ $C = 1$

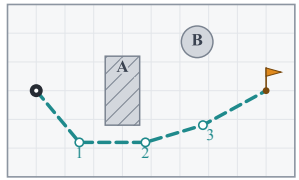
U Upper detour



k	0	1	2	3
a_x	1.5	2.3	2	2.2
a_y	1.8	0	0.7	−2.5

$E = 26.36$ $C = 0$

L Lower detour — selected

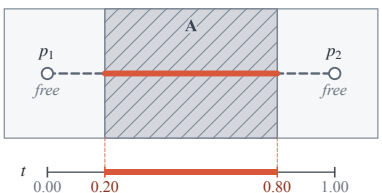


k	0	1	2	3
a_x	**1.5**	2.3	2	2.2
a_y	**−1.8**	0	0.6	1.2

$E = 21.42$ $C = 0$

(c) Swept collision test

D, segment $k = 1$: (3, 3) → (5, 3)

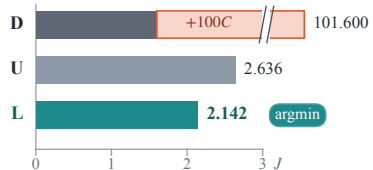


inside A for $t \in [0.20, 0.80] \Rightarrow C = 1$

(d) Cost per candidate

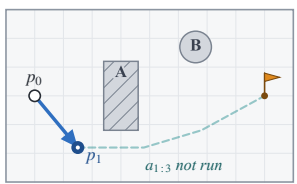
$J = \|p_4 - g\|_2^2 + 0.1E + 100C$

$\|p_4 - g\|_2^2 = 0$ *for all three*



(e) Executed first step

only a_0 of L is applied



$a_0 = (1.5, -1.8)$

$p_1 = (2.5, 1.2)$

swept: free

$o_1 = R(s_1)$

replan